Document 0 — Architecture Index & Reading Guide

Project: Digital Harmonium Workstation

Document Version: 1.0

Purpose: Master Index for the Complete Engineering Architecture

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Purpose of This Document

This document serves as the entry point to the entire Digital Harmonium Workstation architecture.

Rather than containing technical implementation details, it explains:

- What each design meeting covers.
- Where specific engineering decisions are documented.
- How the software architecture is organized.
- The recommended reading order for future developers and AI assistants.

Every engineer or AI working on this project should begin with this document before reading the remaining architecture documents.

Overall Architecture Philosophy

The workstation has been designed using a **top-down engineering approach**.

The project was intentionally designed before implementation.

Every subsystem, task, memory region, runtime state, and data flow has been specified before writing firmware.

The implementation phase should therefore follow the architecture rather than redesign it.

Meeting Index

Meeting 1 — Project Vision

Defines the identity of the workstation.

Focus:

- Project goals
- Design philosophy
- Target users
- Overall product direction

Meeting 2 — Feature Definition

Defines every major feature included in the workstation.

Focus:

- Instrument capabilities
- Recording
- Playback
- Accompaniment
- User features

Meeting 3 — User Interface

Defines how musicians interact with the workstation.

Focus:

- LCD screens
- Menu system
- Navigation
- Button behavior
- Status indicators

Meeting 4 — Runtime Architecture

Defines how the firmware behaves during execution.

Focus:

- Runtime states
- System states
- UI states
- Recording states
- Playback states
- Background subsystem states

Meeting 5 — Software Components

Defines every software module in the firmware.

Focus:

- Module responsibilities
- Software decomposition
- System organization

Meeting 6 — FreeRTOS Tasks

Defines execution architecture.

Focus:

- Tasks
- Priorities
- Scheduling
- DMA
- Interrupts
- CPU execution model

Meeting 7 — Timing Architecture

Defines timing behavior.

Focus:

- Task frequencies
- Refresh rates
- Audio timing
- Keyboard scanning

- Background timing

Meeting 8 — Memory Architecture

Defines Flash and RAM usage.

Focus:

- RAM allocation
- Flash allocation
- Buffer strategy
- Instrument storage
- ESP32 resource utilization

Meeting 9 — Data Flow Architecture

Defines how information moves through the workstation.

Focus:

- Audio flow
- Recording flow
- Playback flow
- Instrument loading
- Control flow
- Internal communication

Meeting 10 — Failure Handling

Defines fault tolerance.

Focus:

- Error recovery
- Runtime protection
- Audio protection
- Memory failures
- Storage failures
- Recovery philosophy

Software Architecture Layers

The firmware is organized into seven logical layers.

Hardware Layer

Interfaces directly with physical hardware.

Examples:

- GPIO
- I²S
- SD Card
- Display
- Timers

Runtime Layer

Controls execution of the entire firmware.

Examples:

- Scheduler
- Runtime Manager
- Task Manager
- System Manager

Instrument & Audio Layer

Generates musical sound.

Examples:

- Voice Manager
- Audio Engine
- Mixer
- Envelope Engine
- Instrument Manager
- Sample Engine
- Tabla Engine
- Tanpura Engine

Input Layer

Processes musician interaction.

Examples:

- Keyboard Scanner
- Button Manager
- Slider Manager
- UI Navigation

Feature Layer

Implements workstation functionality.

Examples:

- Recorder
- Playback
- Presets
- Settings
- Storage Manager

Storage Layer

Manages persistent data.

Examples:

- Instrument installation
- Settings
- Recordings
- Flash management

Diagnostic Layer

Monitors system health.

Examples:

- Logger

- Diagnostics
- Error reporting
- Performance monitoring

Recommended Reading Order

For new engineers or AI assistants:

1. Document 0 — Architecture Index
2. Meeting 1 — Project Vision
3. Meeting 2 — Feature Definition
4. Meetings 3–10 in numerical order
5. Implementation Roadmap
6. Source Code

Following this order provides the necessary context before implementation begins.

Implementation Rule

The architecture documents define the intended behavior of the workstation.

Implementation should conform to these documents rather than modify them.

If new requirements arise, the architecture should be updated first, followed by the firmware implementation.

End of Document 0 — Architecture Index & Reading Guide